Final exam · Unit 6 · Heat and wave equations by separation · 2.5 points · 50 min · Standard exam level

P22 — Heat equation in a rod with insulated ends: conservation of the total heat, the zero eigenvalue, a cosine-power initial profile and the time for the midpoint to settle

A homogeneous metal rod of length $L = 0.5$ m, with lateral surface and both ends thermally insulated, has thermal diffusivity $k = 1.0 \times 10^{-4}$ m²/s. Its temperature $u(x,t)$ (in °C) obeys

$$u_t = k\,u_{xx}, \quad 0 < x < L,\ t > 0, \qquad u_x(0,t) = u_x(L,t) = 0, \qquad u(x,0) = 20 + 40\cos^4\left(\frac{\pi x}{2L}\right).$$

(a) Without solving the equation, prove that the total heat $H(t) = \displaystyle\int_0^L u(x,t)\,dx$ (proportional to the thermal energy, with constant $c\rho A$) does not change in time. Deduce the uniform temperature u_∞ the rod tends to, assuming it tends to a uniform temperature.

(b) Solve the problem by separation of variables. Discuss the three signs of the separation constant, including the zero eigenvalue, and write the initial profile as a finite cosine combination using a power-reduction identity.

(c) Confirm u_∞ from the solution. Find the time constant of the slowest decaying mode and the time t_{10} after which the deviation of the midpoint temperature $u(L/2,t)$ from u_∞ has fallen to one tenth of its initial value. Give the temperatures of both ends at t_{10}.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $u(x,t)$ (°C), independent variables $x \in [0, L]$ (m) and $t \geq 0$ (s); u is C^2 in x and C^1 in t for $t > 0$.
- Linear, homogeneous, second-order parabolic PDE with constant coefficient $k > 0$; homogeneous Neumann (insulated) conditions at both ends (Haberman Sec. 1.3); nonhomogeneous initial condition, smooth and compatible with the end conditions ($f'(0) = f'(L) = 0$).
- No heat sources and no flux through the boundary.

Symbol	Value	Unit
L	0.5	m
k	1.0×10^{-4}	m²/s
$f(x) = u(x,0)$	$20 + 40\cos^4\frac{\pi x}{2L}$	°C
$f(0)$, $f(L/2)$, $f(L)$	60, 30, 20	°C

Phase 2. Canonical formulation

Product solutions $u = X(x)T(t)$ lead to the Sturm-Liouville problem $X'' + \lambda X = 0$, $X'(0) = X'(L) = 0$ ($p = \sigma = 1$, $q = 0$) and to $T' = -\lambda k T$. The initial profile is rewritten as

$$f(x) = A_0 + \sum_{n \geq 1} A_n \cos \frac{n\pi x}{L}.$$

Part (a) · 0.6 points

Conservation of the total heat and the final temperature

Step 1. Derivative of the total heat

The limits 0 and L do not depend on t and u_t is continuous, so we may differentiate under the integral (Haberman Sec. 1.2):

$$H'(t) = \frac{d}{dt} \int_0^L u(x,t)\, dx = \int_0^L u_t(x,t)\, dx = k \int_0^L u_{xx}(x,t)\, dx.$$

An antiderivative of u_{xx} with respect to x is u_x, so by Barrow's rule

$$H'(t) = k \big[u_x(x,t) \big]_{x=0}^{x=L} = k\big(u_x(L,t) - u_x(0,t) \big) = k(0 - 0) = 0.$$

Hence $H(t) = H(0)$ for all t: insulated ends let no heat in or out (Haberman Sec. 1.4).

Step 2. Initial total heat

With $\cos^4\theta = \frac{3 + 4\cos 2\theta + \cos 4\theta}{8}$ (proved in (b), Step 4) and $\theta = \frac{\pi x}{2L}$, $f = 35 + 20\cos\frac{\pi x}{L} + 5\cos\frac{2\pi x}{L}$. For $n \geq 1$, with $s = \frac{n\pi x}{L}$, $ds = \frac{n\pi}{L} dx$:

$$\int_0^L \cos \frac{n\pi x}{L}\, dx = \frac{L}{n\pi} \Big[\sin s \Big]_0^{n\pi} = \frac{L}{n\pi} \big(\sin n\pi - \sin 0 \big) = 0.$$

So $H(0) = \int_0^L 35\, dx = 35L$.

Step 3. Final temperature

If $u \to u_\infty$ (a constant) as $t \to \infty$, then $H \to u_\infty L$; but $H = 35L$ for all t, so

$$u_\infty = \frac{1}{L}\int_0^L f(x)\,dx = 35\ ^\circ\mathrm{C},$$

the mean of the initial temperature, not the mean $40\ ^\circ\mathrm{C}$ of the end values 60 and 20.

KEY RESULTS

- $H'(t) = k\big[u_x(L,t) - u_x(0,t)\big] = 0$, so $H(t) = H(0)$.
- $H(0) = \int_0^L f\,dx = 35L$, so $u_\infty = \dfrac{1}{L}\int_0^L f\,dx = 35\ ^\circ\mathrm{C}$.

Part (b) · 1.3 points

Separation of variables and the solution

Step 1. Separation

With $u = X(x)T(t)$: $XT' = kX''T$; dividing by kXT,

$$\frac{T'}{kT} = \frac{X''}{X} = -\lambda.$$

The end conditions give $X'(0)T(t) = 0$ and $X'(L)T(t) = 0$, so $X'(0) = X'(L) = 0$ for a nontrivial T.

Step 2. The three cases for $X'' + \lambda X = 0$

- $\lambda = -\mu^2 < 0$: $X = A\cosh\mu x + B\sinh\mu x$, $X' = \mu(A\sinh\mu x + B\cosh\mu x)$. $X'(0) = \mu B = 0 \Rightarrow B = 0$. $X'(L) = \mu A\sinh\mu L = 0$ with $\sinh\mu L > 0$, so $A = 0$: trivial.
- $\lambda = 0$: $X = A + Bx$, $X' = B$; $X'(0) = B = 0$ and $X'(L) = 0$ holds for any A. Eigenvalue $\lambda_0 = 0$ with $X_0 = 1$: the constant temperature is an equilibrium of the insulated rod (Haberman Sec. 2.4).
- $\lambda = \mu^2 > 0$: $X = A\cos\mu x + B\sin\mu x$, $X' = \mu(-A\sin\mu x + B\cos\mu x)$. $X'(0) = \mu B = 0 \Rightarrow B = 0$. $X'(L) = -\mu A\sin\mu L = 0$ with $A \neq 0$, so $\sin\mu L = 0$, $\mu = \frac{n\pi}{L}$: $\lambda_n = \left(\frac{n\pi}{L}\right)^2$, $X_n = \cos\frac{n\pi x}{L}$, $n = 1, 2, \ldots$

Step 3. Time factors

$T' = -\lambda_n k T$ is separable: for $T \neq 0$, $\frac{dT}{T} = -\lambda_n k\, dt$, integrate from 0 to t with Barrow's rule, $\ln|T(t)| - \ln|T(0)| = -\lambda_n k t$, so $T_n(t) = T_n(0) e^{-\lambda_n k t}$. For $n = 0$, T_0 is constant. By superposition (Haberman Sec. 2.2):

$$u(x,t) = A_0 + \sum_{n=1}^{\infty} A_n\, e^{-k(n\pi/L)^2 t} \cos\frac{n\pi x}{L}.$$

Step 4. Initial profile as a finite cosine sum

With $\cos^2\theta = \frac{1+\cos 2\theta}{2}$ applied twice:

$$\cos^4\theta = \Big(\frac{1+\cos 2\theta}{2}\Big)^2 = \frac{1 + 2\cos 2\theta + \cos^2 2\theta}{4} = \frac{1}{4}\Big(1 + 2\cos 2\theta + \frac{1+\cos 4\theta}{2}\Big) = \frac{3 + 4\cos 2\theta + \cos 4\theta}{8}.$$

With $\theta = \frac{\pi x}{2L}$: $2\theta = \frac{\pi x}{L}$, $4\theta = \frac{2\pi x}{L}$, and

$$f(x) = 20 + 40 \cdot \frac{3 + 4\cos\frac{\pi x}{L} + \cos\frac{2\pi x}{L}}{8} = 20 + 15 + 20\cos\frac{\pi x}{L} + 5\cos\frac{2\pi x}{L}.$$

Check: $f(0) = 35 + 20 + 5 = 60$, $f(L/2) = 35 + 0 - 5 = 30$, $f(L) = 35 - 20 + 5 = 20$, as in the data table.

Step 5. Coefficients and solution

f is already a combination of eigenfunctions, so matching $u(x,0) = f$ term by term (orthogonality of $\cos\frac{n\pi x}{L}$ on $[0,L]$, Haberman Sec. 2.4: $A_0 = \frac{1}{L}\int_0^L f$, $A_n = \frac{2}{L}\int_0^L f\cos\frac{n\pi x}{L}$) gives $A_0 = 35$, $A_1 = 20$, $A_2 = 5$ and $A_n = 0$ for $n \geq 3$:

$$u(x,t) = 35 + 20\, e^{-k\pi^2 t/L^2}\cos\frac{\pi x}{L} + 5\, e^{-4k\pi^2 t/L^2}\cos\frac{2\pi x}{L}.$$

Verification: $u_t = -\frac{20k\pi^2}{L^2} e^{-k\pi^2 t/L^2}\cos\frac{\pi x}{L} - \frac{20k\pi^2}{L^2} e^{-4k\pi^2 t/L^2}\cos\frac{2\pi x}{L}$ and, by the chain rule, $k u_{xx} = k\Big[-20\frac{\pi^2}{L^2} e^{-k\pi^2 t/L^2}\cos\frac{\pi x}{L} - 5\frac{4\pi^2}{L^2} e^{-4k\pi^2 t/L^2}\cos\frac{2\pi x}{L}\Big]$, the same expression. u_x contains only $\sin\frac{\pi x}{L}$ and $\sin\frac{2\pi x}{L}$, which vanish at $x = 0$ and $x = L$.

KEY RESULTS

- $\lambda < 0$: trivial; $\lambda_0 = 0$: $X_0 = 1$; $\lambda_n = \left(\frac{n\pi}{L}\right)^2$: $X_n = \cos\frac{n\pi x}{L}$, $T_n = e^{-k(n\pi/L)^2 t}$.
- $f(x) = 35 + 20\cos\dfrac{\pi x}{L} + 5\cos\dfrac{2\pi x}{L}$.
- $u(x,t) = 35 + 20\,e^{-k\pi^2 t/L^2}\cos\dfrac{\pi x}{L} + 5\,e^{-4k\pi^2 t/L^2}\cos\dfrac{2\pi x}{L}$.

Part (c) · 0.6 points

Final state and settling times

Step 1. Long-time limit

Both exponents are negative, so as $t \to \infty$ the two modes vanish and $u \to A_0 = 35\ ^\circ\text{C}$, the value predicted in (a) from conservation alone. The zero eigenvalue is the reason: its mode does not decay, and its amplitude is the mean of f.

Step 2. Time constants

Mode n decays as e^{-t/τ_n} with $\tau_n = \dfrac{L^2}{kn^2\pi^2}$. The slowest is $n = 1$:

$$\tau_1 = \frac{(0.5)^2}{(1.0\times10^{-4})\pi^2} = \frac{0.25}{9.870\times10^{-4}} \approx 253.3\ \text{s}, \qquad \tau_2 = \frac{\tau_1}{4} \approx 63.3\ \text{s}.$$

Step 3. Midpoint

At $x = L/2$: $\cos\frac{\pi}{2} = 0$ and $\cos\pi = -1$, so $u(L/2,t) = 35 - 5e^{-4k\pi^2 t/L^2}$. The midpoint starts at $30\ ^\circ$C and warms monotonically to $35\ ^\circ$C; its deviation $5e^{-4k\pi^2 t/L^2}$ is governed by the faster mode only. One tenth of the initial deviation:

$$e^{-4k\pi^2 t_{10}/L^2} = \frac{1}{10} \iff \frac{4k\pi^2 t_{10}}{L^2} = \ln 10 \iff t_{10} = \frac{L^2\ln 10}{4\pi^2 k} = \frac{0.25\cdot 2.3026}{4\pi^2\cdot 10^{-4}} \approx 145.8\ \text{s}.$$

Step 4. Ends at t_{10}

$t_{10}/\tau_1 = \frac{\ln 10}{4}$, so $e^{-t_{10}/\tau_1} = 10^{-1/4} \approx 0.5623$, and the second mode equals $5 \cdot \frac{1}{10} = 0.5$. At $x = 0$ (both cosines $= 1$): $u = 35 + 20(0.5623) + 0.5 \approx 46.75$ °C. At $x = L$ ($\cos\pi = -1$, $\cos 2\pi = 1$): $u = 35 - 11.25 + 0.5 \approx 24.25$ °C. The ends are still far from equilibrium when the midpoint has almost settled: the midpoint is a node of the slow mode.

KEY RESULTS

- $u \to 35$ °C; slowest time constant $\tau_1 = \dfrac{L^2}{k\pi^2} \approx 253.3$ s ($\tau_2 = \tau_1/4 \approx 63.3$ s).
- $u(L/2, t) = 35 - 5e^{-4k\pi^2 t/L^2}$; $t_{10} = \dfrac{L^2 \ln 10}{4\pi^2 k} \approx 145.8$ s.
- At t_{10}: $u(0) = 35 + 20 \cdot 10^{-1/4} + 0.5 \approx 46.75$ °C, $u(L) \approx 24.25$ °C.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $t \to 0^+$ reproduces f (Step 4 of (b)); $t \to \infty$ gives the mean 35 °C; for $t \gg \tau_2$, $u - 35 \approx 20e^{-t/\tau_1}\cos\frac{\pi x}{L}$, a single half cosine with the hot end at $x = 0$. Doubling L multiplies all times by 4 (diffusive scaling $t \sim L^2/k$).
- **Checks:** dimensions: $k\pi^2 t/L^2$ has (m^2/s)(s)/m^2, dimensionless; $\tau_1 = L^2/(k\pi^2)$ is in s. $H(t) = 35L$ for all t computed directly from the solution (script). The PDE, both Neumann conditions and the initial condition hold (script); the coefficients also follow from the integrals $\frac{1}{L}\int f$ and $\frac{2}{L}\int f\cos\frac{n\pi x}{L}$ with the original $\cos^4$ form.
- **Common mistakes in the exam:** (1) forgetting the $\lambda = 0$ eigenvalue and its constant mode, which gives a rod that cools to 0 °C although no heat can leave; (2) using sines $\sin\frac{n\pi x}{L}$ (the Dirichlet eigenfunctions); (3) taking the final temperature as the average of the end values, 40 °C, instead of the spatial mean 35 °C; (4) computing t_{10} with the slow rate $k\pi^2/L^2$ at the midpoint, where that mode is absent (t would come out four times too large, 583 s).